EDEM Introductory Tutorial

Screw Auger

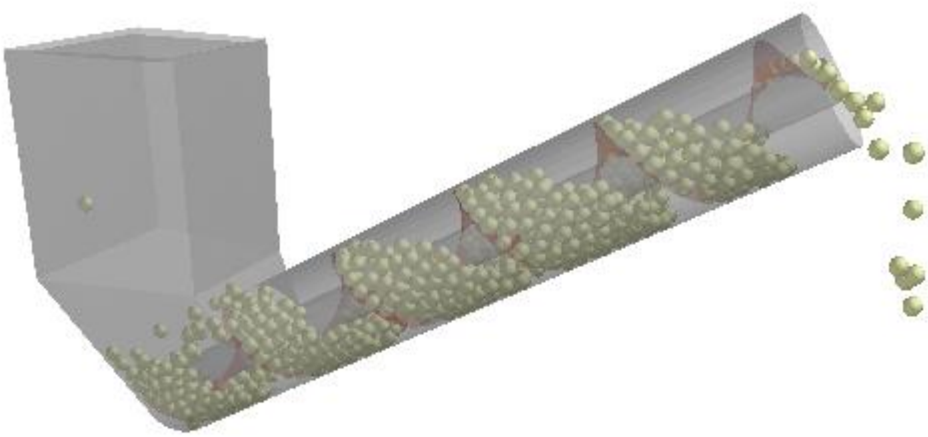
Introduction

This introductory tutorial describes how to set up and analyze an EDEM simulation.

The model presented here is a screw auger, a machine designed to transport material horizontally or up an incline via a rotating screw blade.

The main focus is on:

- Define a domain efficiently to see all the important features in simulation
- Analysis of particles' velocities and contact forces
- Set up data export queries and using the copy query option
- Extraction of the relevant data to the spreadsheet using Step Factor data export
- Creation of a video of the completed simulation



1. Start EDEM and save your project as **Screw_Auger_tutorial** in your desired location (e.g. C:\EDEM_Tutorials).
2. Remember Save often when setting up the simulation.

Please note this tutorial has associated CAD file screw_auger_complete.stl. It is recommended to copy this file to the same location as the simulation files.

EDEM Creator: Setting up the model

Step 1: Define the Project and Settings

Choose the units

The first step in creating the model is to set the units used throughout EDEM.

1. Go to the Tools > Options… and select the Units tab.
2. Change the following measurement units:
   - Angle to degrees (**deg**)
   - Angular velocity to **rpm**
   - Length to **mm**
3. Click OK.

Enter the model title and description

1. Click on Project in the Creator Tree.
2. Enter the title **Screw Auger Model** in the Title box.
3. Enter a description in the Description field.

The model title and description will appear in the Data Browser window.

Step 2: Define the Bulk Material

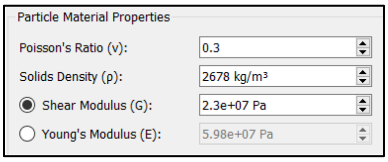
Add new bulk material

The first step is to add the bulk material which will be used in the simulation.

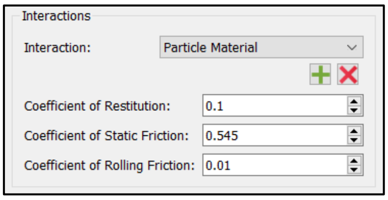
1. Right click on the Bulk Material in the Creator Tree, select Add Bulk Material and enter the name **Particle Material** into the highlighted field (BulkMaterial1). Alternatively, you can also use the icons in the Toolbar.

Define the bulk material properties and interactions

1. Select **Particle Material** under the Bulk Material section.
2. Set Poisson's Ratio, Solid Density and Shear Modulus as shown.
3. Click the  button in the Interaction section and select **Particle Material** when prompted. This will define the interaction between all elements made of Particle Material.



4. Set the coefficients as follows:



Create a particle shape

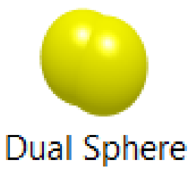
After the material and interaction properties have been defined, the particle shapes for the bulk material need to be created.

1. Right click on the **Particle Material** under the Bulk Material section, select Add Particle > Add MultiSphere and enter the name **Grain Particle** into the highlighted field.

Define the spheres and properties

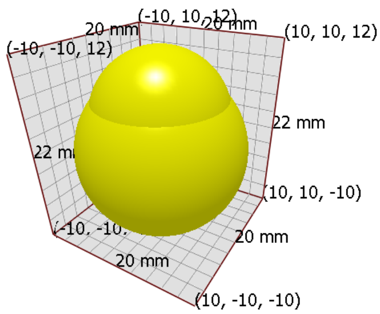
Particles are made up of one or more spheres. In this case the grain particle is to be made from two spheres.

1. Select the **Grain particle** in the Creator Tree.
2. Click on **Modify Shape** and select the Dual Sphere from the Shape Library:



3. In the particles detail table, change the Physical Radius of ‘sphere 0’ to **10 mm** and the Position X, Y, Z to (**0, 0, 0**) mm.

4. Set the Physical Radius of 'sphere 1' to **8 mm** and the Position X, Y, Z to (**0, 0, 4**) mm.



5. Under Grain particle > Properties click the Calculate Properties button to define the mass, volume and inertia.

Set the size distribution of the bulk material

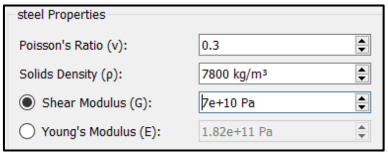
1. In the Creator Tree select Grain particle > Size distribution.
2. Select **normal** from the drop-down list and set the Mean to **1** and the Standard Deviation to **0.05**.
3. Select **Radius** in the Scale By section.
    - Note this means the Mean is 1x particle diameter and the Standard Deviation is 0.05 x particle diameter

This creates a bulk material consisting of a 2-sphere shape with a normal size distribution.

Step 3: Define the Equipment Material

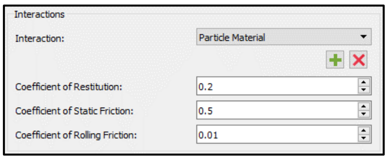
Add new equipment material

1. Right click on the Equipment Material in the Creator Tree, select Add Equipment Material and enter the name **steel** into the highlighted field (EquipmentMaterial1).
2. Select **steel** under the Equipment Material section and set the steel properties as shown.



Define the bulk material - equipment material interaction

1. Select **steel** under the Equipment Material section.
2. Click the  button in the Interaction section. Select **Particle Material** when prompted. This will define the interaction between elements made of Particle Material and elements made of steel.
3. Set the coefficients as follows:



Step 4: Define the Geometries

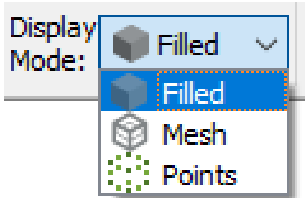
The next step is to define the screw auger geometry used in the model.

Import the conveyor geometry

The screw auger geometry has already been created in a CAD package and is ready to be imported into EDEM.

1. Right click on the Geometries section in the Creator Tree.
2. Select Import Geometry and Navigate to the file **screw_auger_complete.stl**
3. When prompted, set the Units of measurement to **Millimeters**. Leave the Merge Sections option **unchecked** and leave all the settings as the default values and click OK.

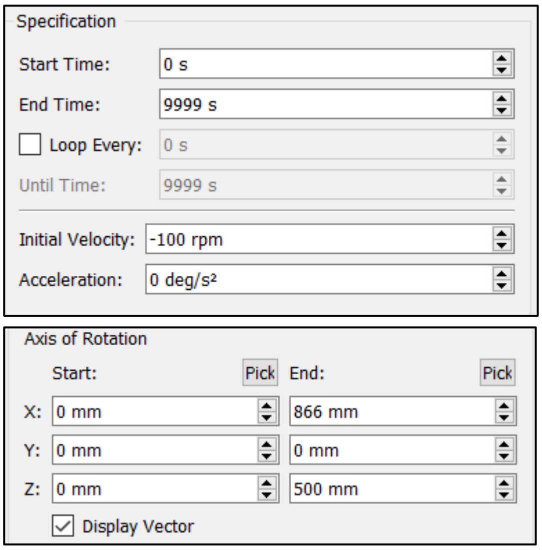
The screw auger will appear in the Viewer window. It is made up of two sections which are listed under the Geometries section in the Creator Tree. When any are selected, they are highlighted in red.



4. In the top toolbar Select **Mesh** as the Geometry Display Mode.
5. Right click each geometry element in the Geometries section and rename them to ***hopper*** *and* ***screw blade*** as appropriate.
6. Holding down the Ctrl or Shift key, select all the geometry sections. Set their Type to **Physical** and change the Material to **steel** for all of them.

Specify the screw kinematics

In this model the screw blade rotates in order to lift the particles. This rotation is set-up by specifying a linear rotation kinematic for the screw section.

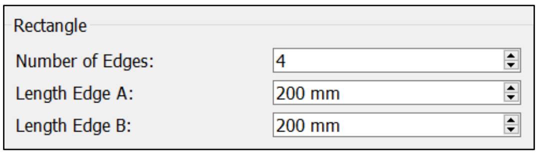


1. Right click on the **screw blade** in the Creator Tree.
2. Select Add Motion and then choose Add Linear Rotation Kinematic.
3. Select the newly created Kinematic and set the Initial Velocity and the Axis of Rotation as shown:
4. As the **Display Vector** box is checked, you can see the axis of rotation and direction in the Viewer window.

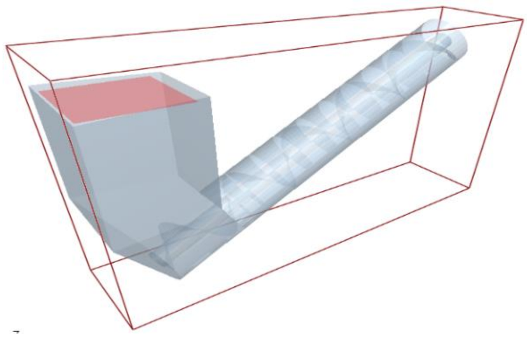
Create the particle factory plate

Particle factories are used to define where, when and how particles appear in a simulation. All factories must be based on a section of geometry (whether 'real' or 'virtual'). This defines the area or volume in the model that produces the particles.

1. Right click on the Geometries section in the Creator Tree.
2. Select Add Geometry, then **Polygon** and enter the name **factory_plate** into the highlighted field.
3. Select the factory_plate and in the General section of the Detailed View set the type to **Virtual**, since the plate is not a physical part of the machinery.
4. Expand the factory_plate section, select the Transform subsection and set the Z position to 480 mm.
5. Expand the factory_plate section, select the Polygon subsection and set the size as follows:



Notice that the factory plate is created at the top of the hopper.



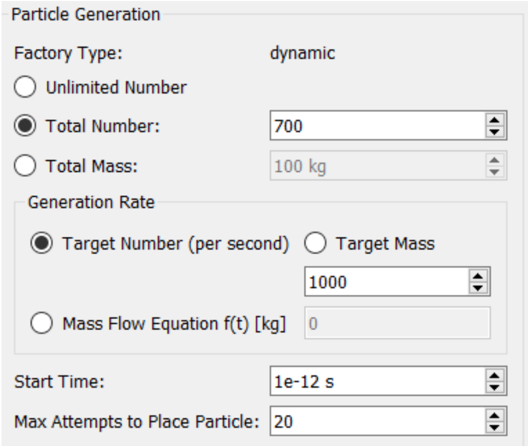
Create the particle factory

1. Right click on the factory_plate section and select Add Factory.

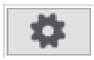
Set the factory’s particle creation values

1. Select the newly created factory and make sure the Factory Type is set to **dynamic**. If it is not, right click on the new factory in the Creator Tree and click Change Factory Type.

2. Set the **Total Number** of particles produced and **Generation Rate** as follows:



Set the factory's initial parameters

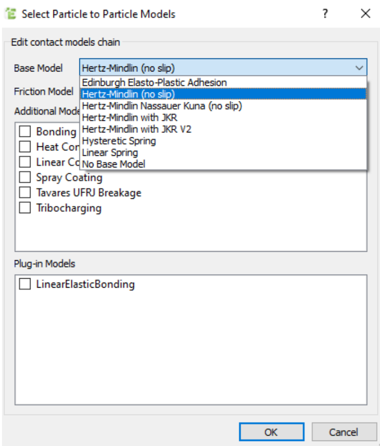
1. Set the Velocity to **fixed** and click the configure button  to set the velocity in z-direction to **-2 m/s**. In this model, you want the particles to be traveling downwards at this speed towards the bottom of the hopper.
2. Select File > Save.

Step 5: Define the Physics

Set the Contact Models

The Physics section lists contact models that describe how elements behave when they come into contact with one another. In this example, the default Hertz-Mindlin contact model and Standard Rolling Friction are used for both Particle to Particle and Particle to Geometry interactions.

1. Select the Physics section and ensure that the **Hertz-Mindlin (no slip)** model is selected for both Particle to Particle and Particle to Geometry interactions.
2. Select the Standard Rolling Friction model. If the models have to be changed click on Edit Contact Chain and select the required contact models.

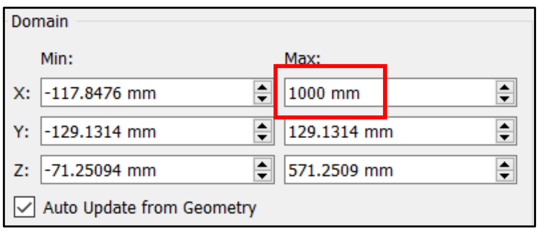


Step 6: Define the Environment

Set the domain of the simulation

The domain (illustrated by the red box) is the area in which simulation will take place. The EDEM Simulator stops tracking and deletes any particles that move out of the domain during the course of the simulation.

1. In the Creator Tree, select the Environment section.
2. Uncheck the **Auto Update from Geometry** box in the Domain section and increase the Max domain to 1000mm in the X-direction. This will enable you to see the particles falling from the end of the screw.



EDEM Simulator: Running a Simulation

Once the model has been set-up in the Creator we move to the Simulator to run it.

1. Click on the Simulator button  on the Toolbar.

Step 1: Set the Time options

Set the time step

The time step is the amount of time between iterations (calculations) in the Simulator.

1. Leave the Auto Time Step box ticked and leave the default Time Integration method as Euler.

Set the simulation time and data write-out frequency

The simulation time is the amount of real time your simulation represents.

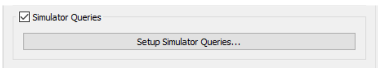
1. In the Simulation Time section, set the Total Time to **10s**.
2. Set the Target Save Interval to **0.01s** to specify the write-out frequency.

This writes out a full EDEM save point and all the particle information every 0.01s. For large simulations this can result in very large file sizes so it is important to consider how often data is required. Saving data every 0.01 s allows a video to be created at 100 Hz when the simulation is complete.

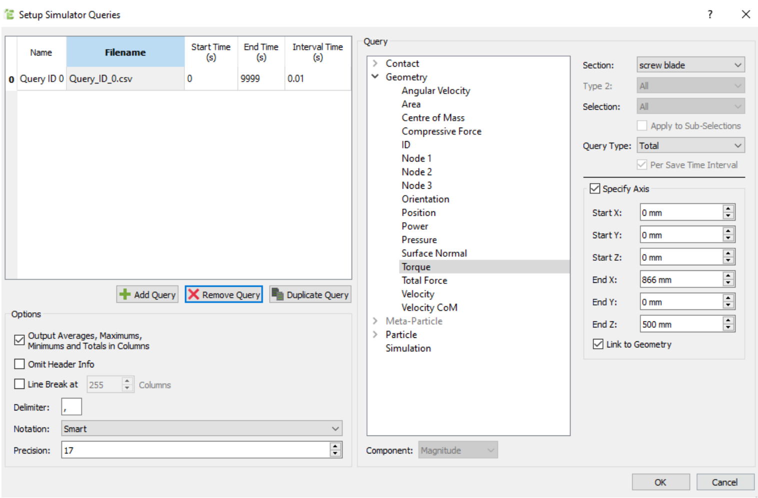
Step 2: Setup Simulator Queries

The previous step outlines the full save intervals (every 0.01 s), we can also export data during simulation at a higher frequency than the standard save. As we are only saving the data required rather than a full save this does not use up as much file space, however if we want to edit this query the simulation has to be re-run or the data exported from the 0.01 s save points.

1. Select 'Simulator Queries' and select 'Setup Simulator Queries'



2. Choose 'Add Query and choose Geometry > Torque > Section: screw blade > Query Type : Total
    a. Ensure Interval Time (s) is set to 0.001 s
    b. Select Specify Axis and set the same end point as previously defined with the kinematics

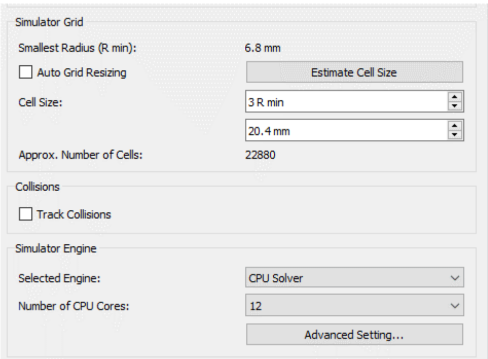


Step 3: Set the Grid options (CPU Engine Only)

1. The grid options are used to optimize the simulation. Set the Cell Size to **3 Rmin.**

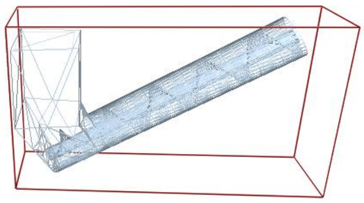
The CPU Solver uses a grid to optimize the particle contact detection. This does not influence the results only the speed of the solver. Typically 2-3 Rmin is optimal with larger values required for larger particle size distrbutions.

The GPU CUDA solver does not use the grid option, the domain is automatically split using a Bounding Volume Hierarchy (BVH) type method, this does not require any user input.

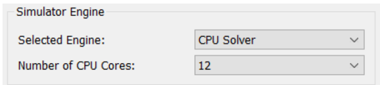


Step 4: Run the Simulation

1. In the Toolbar, set the Display Mode to **Mesh** and Opacity to **0.7** to more clearly see what is happening in your simulation.



2. Select the appropriate number of CPU cores for your simulation. The total number available depends on your license and hardware. This simulation can also be run using the GPU Engine which will typically be 10-100 times faster than the CPU solver, however the GPU solver is better suited for simulations with greater numbers of particles.



3. Click the start Progress button  at the bottom of the viewer window.

4. In Viewer Controls, click the **Refresh Viewer** button  or enable **Auto Update**  at any point to update the Viewer and see how the simulation is progressing.

5. Note that during simulation Query_ID_0.csv will be written to the same folder as the simulation files are saved to.
    a. The torque query will be automatically written to this file and updated during the simulation run.

EDEM Analyst: Analyzing Your Results

The Analyst is used to review, examine and analyze the results of the simulation.

1. Click on the Analyst button on the Toolbar.
2. Use the controls below the main window to rewind the simulation to the first time-step.
3. Click the button to play through each time step of your simulation.

Step 1: Configuring the Display

The options in the Display section of the Analyst Tree are used to configure how the different elements in your model appear. The particles, geometry and contacts within your model can be colored in a variety of ways. Any section of geometry, particle, contact type or selection group can be colored independently.

Configure the geometry

Geometry sections can be displayed as solid shapes, as meshes or as points (based on the mesh nodes). The opacity can also be altered. These display options can be applied to all of your geometry, or just to individual sections.

1. Expand the Display section in the Analyst Tree and then expand the Geometries subsection.
2. Experiment with altering the geometry display.
3. Before finishing ensure that the hopper is drawn as a **Mesh** with **0.7** Opacity, the screw blade is **Filled**, and the factory plate Display box is unchecked.

Configure the particles

The display of the particles can be altered in a similar way to that of the geometry. In addition, the particles can be represented using a number of different forms, like cones, streams and vector that traces the direction in which particles move.

1. Use the Jump to Start button to set the simulation back to the start.
2. Select the Grain particle subsection in the Analyst Tree. To be able to do this some particles must have been generated in the domain. Press the step forward button once and the Grain particle should be available to select.
3. Set the Representation type to **Vector**.
4. Click the play button and watch how the direction of movement of the particle changes over time.
5. Click the stop button at any point and set the Representation type back to **Default**.

Configure the contacts

You can choose to show all contacts or only a subset of them; for example just contacts between particles, or between particles and a particular section of geometry. Contacts can also be displayed in a number of ways by using the contact vector, tangential force, normal force or normal overlap.

1. In the Contact section of the Analyst Tree, experiment with turning on and off the **Display** of contacts between elements of different types. It may be useful to temporarily turn off the display of particles to better see the contacts.
2. Before you finish, turn off the **Display** of all contacts and make sure that the particles **Display** is enabled.

Color the geometry

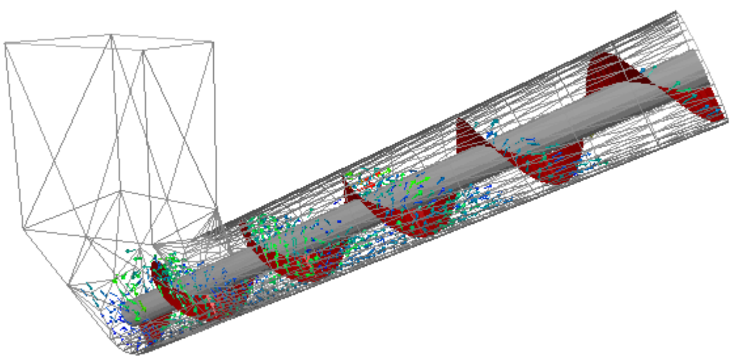
Color the screw blade separately from the rest of the geometry in order to highlight it.

1. Select the **screw blade** geometry element in the Analyst Tree.
2. In the Coloring section of the Detailed View, select **Uniform** in the Color by pull-down menu.
3. Set the Color to the **Dark Red** to color the blade.
4. Set the Color of the remaining sections of the geometry to **Dark Gray**.

Color the particles according to their velocity

Next, color the particles according to a particular attribute – their velocity.

1. Select **Grain particle** in the Particles subsection of the Analyst Tree.
2. In the Coloring section of the Detailed View, select **Velocity** in the Color by pull-down menu.
3. Select **3** levels of coloring and set the Min, Mid and Max colors to **Blue, Green** and **Red** respectively. Leave the Smooth Colors button checked.
4. Click the  button next to the Min and Max Value fields – this reads the current values from the point in the simulation being displayed in the Viewer at the current moment. Check the **Auto Update** option so that the values continue to update whilst the simulation is being played.
5. Rewind the simulation back to the start and click the Play button and watch how the colors update as the simulation progress.



Step 2: Extracting Data

Data collected during your simulation can be exported for analysis. Data can be exported from a single time step, a range of time steps, from all elements in your model or just from a selection or bin group.

Create an export configuration

1. Select File > Export > Results Data. The Export Results Data dialog will appear.
2. Press the button under Configuration ID to create a configuration. A single configuration can contain multiple queries and can be reused at any point in your model.
3. Set the Export format to **Text** so the data will be extracted to a .csv file.
4. Enter the file name ***Screw Auger*** and set the save location to the tutorial folder.

Define a query

Data has been saved every 0.01 s, exporting a large amount of data can take some time, so in this instance the Export Data dialog can be set to only export data every 0.1 s.

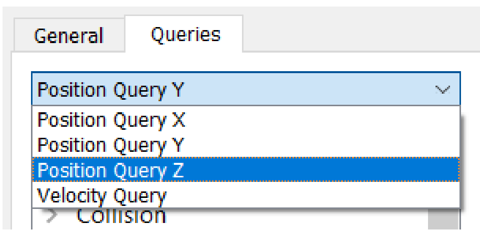
1. In Export Results Data dialog, go to the General tab.
2. Ensure the **Output Data in Columns** checkbox is ticked.
3. Under Time Steps, uncheck the **All** box and set the start time to 0 s and end time to ~10 s.
4. Under Step Factor, choose Time and set the value to 0.1 s.

Add a query to export the average particle velocity

5. Click the Queries tab to open the Queries pane. This is where you define the data to be exported.
6. Click the button to create a new query. Name the query ***Velocity Query***, by pressing the edit name button .
7. Click on the Particle element in the list and then select the attribute **Velocity**.
8. Set the Component to **Magnitude** and the Query type to **Average**. Leave the other options with their default settings.

Add a query to export the X, Y and Z-position of each particle

9. Click the button to create a second query. Name it ***Position Query X***.
10. Click on the Particle element in the list and then select the attribute **Position**.
11. Set the Component to **X** and Query Type to **Average.**
12. *Use the button to copy the query, rename* it to ***Position Query Y*** *and set the component to* ***Y.***
13. *Copy and repeat for the Z-component.*



Export the data

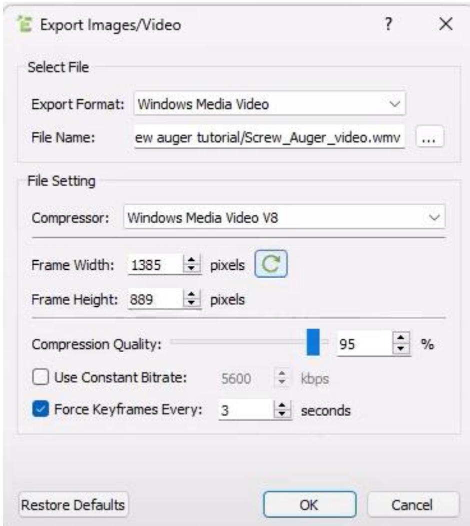
1. Click the Export button to export the data to the pre-defined file.
2. Open the file in an external application (such as a spreadsheet) and review the data.

Step 3: Creating a Video

EDEM can export videos as AVI, WMV, MKV, MP4, WebM, or EnSight Video file formats.

Set video options

1. Click the gray record button  in the bottom right of the viewer to open the Export Video window:



2. Select **Windows Media Video** as the Export Format.
3. Set the File Name and choose a location to save a video, for example save the file as **C:\EDEM_Tutorials\Screw_Auger_video.wmv**
4. Select **Windows Media Video V8** or any available compressor. EDEM will detect the compressor installed on the local machine so this may vary depending on computer specifications.
5. Click the screen grab button  to set the frame width and height to be the same as the dimensions of the viewer.
6. Click OK. The record button turns red: 

Record the video

1. Click the jump-to-start button  (if necessary) then click the play button .
2. Experiment with stopping playback, changing camera angles and display options, then restart playback.

3. Once playback completes, **click the record button again** to save the video file.
4. Open the video using Windows Media Player or any other media player.